

● MEDIUM

● ALGEBRA

Systems of two linear equations in two variables

03

10 questions · ~15 min

Q2

ALGEBRA

$$y = 3x$$

$$2x + y = 12$$

The solution to the given system of equations is x, y . What is the value of $5x$?

- A. 24
- B. 15
- C. 12
- D. 5

Q3

ALGEBRA

$$x + 3 = -2y + 5$$

$$x - 3 = 2y + 7$$

The solution to the given system of equations is x, y . What is the value of $2x$?

- A. -2
- B. 6
- C. 12
- D. 24

The combined original price for a mirror and a vase is \$ 60. After a 25% discount to the mirror and a 45% discount to the vase are applied, the combined sale price for the two items is \$ 39. Which system of equations gives the original price m , in dollars, of the mirror and the original price v , in dollars, of the vase?

- A. $m + v = 60$
 $0.55m + 0.75v = 39$
- B. $m + v = 60$
 $0.45m + 0.25v = 39$
- C. $m + v = 60$
 $0.75m + 0.55v = 39$
- D. $m + v = 60$
 $0.25m + 0.45v = 39$

Q5**ALGEBRA**

A wire with a length of 106 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 6 more than 4 times the value of y . What is the value of x ?

- A. 25
- B. 28
- C. 56
- D. 86

Q6**ALGEBRA**

$$y = -\frac{1}{5}x$$

$$y = \frac{1}{7}x$$

The solution to the given system of equations is x, y . What is the value of x ?

- A. -5
- B. 0
- C. 2
- D. 7

Q7**ALGEBRA**

$$3y = 4x + 17$$

$$-3y = 9x - 23$$

The solution to the given system of equations is x, y . What is the value of $39x$?

- A. -18
- B. -6
- C. 6
- D. 18

Q8**ALGEBRA**

$$y - 9x = 13$$

$$5x = 2y$$

What is the solution x, y to the given system of equations?

- A. $\frac{5}{2}, 1$
- B. $1, \frac{2}{5}$
- C. -2, -5
- D. -5, -2

Which of the following systems of linear equations has no solution?

A. $x = 3$
 $y = 5$

B. $y = 6x + 6$
 $y = 5x + 6$

C. $y = 16x + 3$
 $y = 16x + 19$

D. $y = 5$
 $y = 5x + 5$

Q10

GRID-IN ALGEBRA

$$5x + 3y = 38$$

$$x + 3y = 10$$

In the solution (x, y) to the system of equations above, what is the value of x ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.